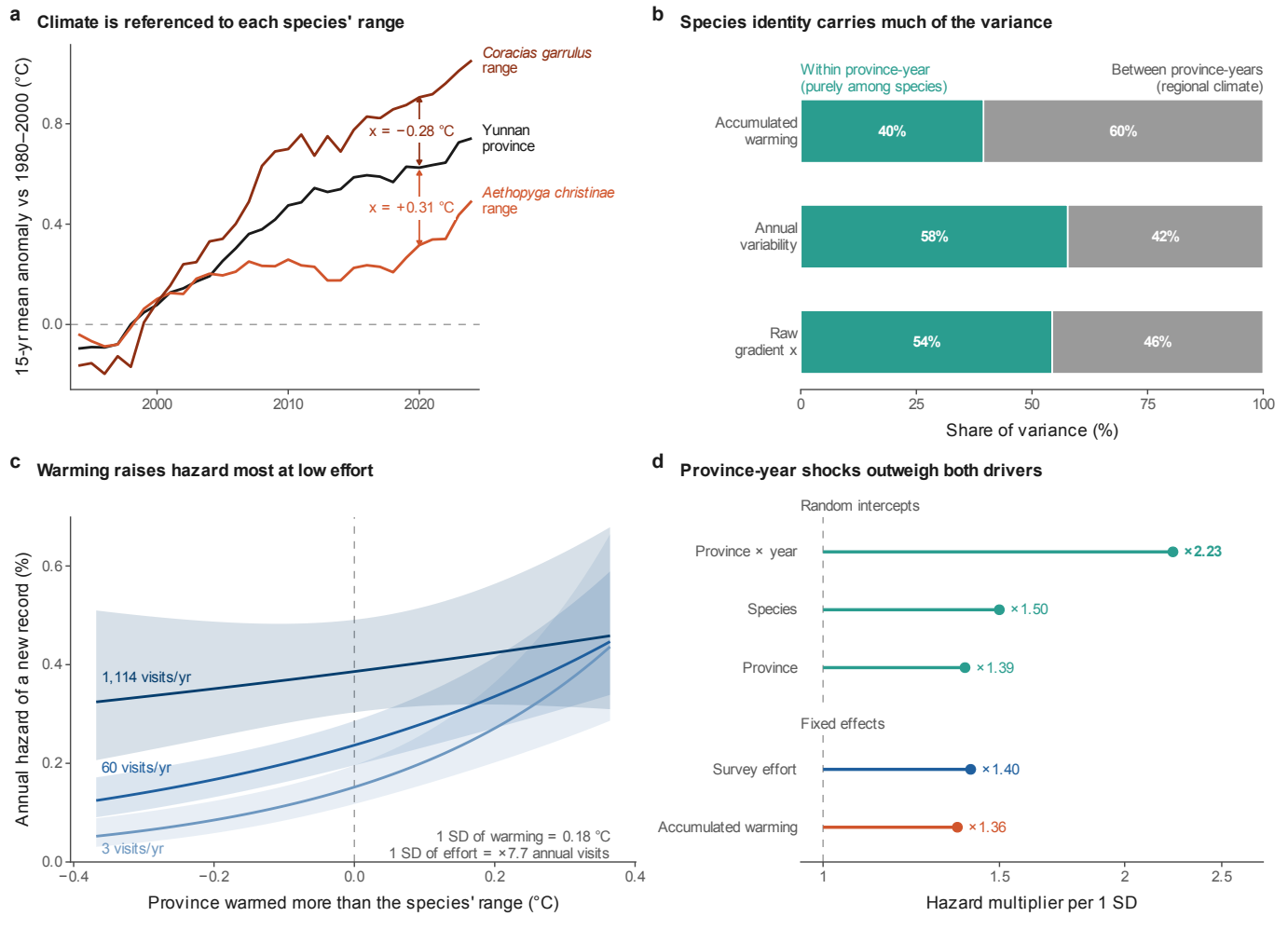

a
Climate is referenced to each species' range
15-yr mean anomaly vs 1980–2000 (°C)
0.0
0.4
0.8
2000
2010
2020
Coracias garrulus range
Yunnan province
Aethopyga christinae range
x = −0.28 °C
x = +0.31 °C
b
Species identity carries much of the variance
Within province-year (purely among species)
Between province-years (regional climate)
Accumulated warming
Annual variability
Raw gradient x
40%
60%
58%
42%
54%
46%
0
25
50
75
100
Share of variance (%)
c
Warming raises hazard most at low effort
Annual hazard of a new record (%)
0.0
0.2
0.4
0.6
1,114 visits/yr
60 visits/yr
3 visits/yr
1 SD of warming = 0.18 °C
1 SD of effort = ×7.7 annual visits
−0.4
−0.2
0.0
0.2
0.4
Province warmed more than the species' range (°C)
d
Province-year shocks outweigh both drivers
Random intercepts
Province × year
×2.23
Species
×1.50
Province
×1.39
Fixed effects
Survey effort
×1.40
Accumulated warming
×1.36
1
1.5
2
2.5
Hazard multiplier per 1 SD